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ICRR Fellow

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Research interests

Galaxy evolution; the galaxy–IGM connection; quasars and supermassive black holes; protoclusters and large-scale structure; high-redshift galaxies and cosmic reionization.

Appointments

- **2025-10–present · ICRR Fellow**
ICRR, The University of Tokyo
- **2024-10–2025-09 · Project Researcher**
National Astronomical Observatory of Japan
- **2022-10–2024-09 · Project Researcher**
ICRR, The University of Tokyo
- **Jan–Feb 2024 · Visiting Researcher**
University of Milano-Bicocca; host: Sebastiano Cantalupo

Education

- **2017–2022 · Ph.D., SOKENDAI**
Astronomy research based at NAOJ. Thesis: *Correlation between galaxy and IGM H I at $z \approx 2$ mapped by Subaru/HSC*. Supervisors: Nobunari Kashikawa and Masayuki Tanaka.
- **2013–2017 · Bachelor's degree in astronomy**
University of Science and Technology of China. Supervisor: Junxian Wang.

Research funding

- **2024–2027 · Principal Investigator · JSPS KAKENHI 24K17084**
Early-Career Scientists. Elucidating the Trigger and Feedback Mechanisms of the Most Concentrated Quasars at Cosmic Noon. JPY 4,550,000 including indirect costs; JPY 3,500,000 direct.
- **2025–2030 · Co-Investigator · JSPS KAKENHI 25H00663**
Scientific Research (A). Galaxy Overdense Regions in the Distant Universe Revealed by Wide-Area Sky Survey and TAO. PI: Nobunari Kashikawa.

Professional service & membership

- JCMT Time Allocation Committee, September 2023–November 2025.
- International Astronomical Union member, since June 2024.
- Astronomical Society of Japan member, since October 2017.

Observing & research programs

50 entries, by selection year or cycle start, newest first. Roles refer to my participation. Joint facilities and repeat semesters are listed separately. ALMA Grade C entries are conditional; requested and observed times are explicitly labeled. Awards do not imply completed observations.

Chandra · Associated U.S. observing support

Science PI: Yongming Liang · 340 ks awarded · Cycle 27, proposal 27700187.

US\$61,320 approved in total, including Co-I grants. U.S. administrative PI: **Martin Elvis**. This supports the U.S. investigators on the project and is administered on the U.S. side.

Chandra funding policy · Non-U.S. science PI and U.S. administrative PI have distinct roles.

2026

PI

Cycle 13 2026.1.01512.S

ALMA / Band 3

Approved program

Molecular Reservoirs in Cosmic Himalayas: A Uniform Survey of $z \sim 2.2$ CO(3-2) Emitters in the Densest QSO Group

PI: Yongming Liang · East Asia

Grade B. Approved for Cycle 13.

[Program details ↗](#)

Co-I

Cycle 5 JWST-10005

JWST / NIRSpec, NIRISS & NIRCам

34.1 h awarded

Little Red Dots in a Time (Delay) Machine: Monitoring Gravitationally Lensed, Multiply Imaged LRDs over 160 years

PI: Fengwu Sun

[Program details ↗](#)

Co-I

Cycle 5 JWST-11371

JWST / NIRSpec

108.5 h awarded

Mapping Reionisation in 3D in the COSMOS field

PI: Romain Meyer

[Program details ↗](#)

Co-I

Cycle 5 JWST-9483

JWST / MIRI

18.7 h awarded

Pushing Little Red Dots to the Early Universe: Spectroscopic Confirmation of the First Robust $z \sim 10$ LRD Candidate

PI: Takumi Tanaka

[Program details ↗](#)

PI S26B S26B-081

Keck / KCWI

0.5 nights awarded

Origin of Ly α Nebulae: Extreme Test in the Densest $z>2$ Quasar Overdensity

PI: Yongming Liang · Subaru time exchange

[Program details ↗](#)

Co-I Cycle 1 Roman-19061

Roman

Analysis program

Building Roman's Infant Galaxies with Hyper-refined Targeted FIRE Simulations

PI: Guochao Sun

Bright-FIRE: theory and simulation program; no new observing-time allocation.

[Program details ↗](#)

Co-I S26A S26A-065

Subaru / HSC

1 night awarded

Mapping gas and galaxies in proto-supercluster at $z=2.23$ with QSO clustering

PI: Ko Ishida

[Program details ↗](#)

PI S26A S26A-067

Subaru / MOIRCS

0.5 nights awarded

Emission-line Mapping in a Protocluster at $z=3$ Fed by Inspirling Cold Streams

PI: Yongming Liang · Service program

[Program details ↗](#)

Co-I S26A S26A-131

Subaru / PFS-Q

1.56 nights awarded

3D Tomographic Mapping of Reionization with Ly α Emission in COSMOS-3D

PI: Koki Kakiichi

Queue allocation; 2.23 nights including queue overheads.

[Program details ↗](#)

Co-I S26A S26A-147

Subaru / HSC

4 nights awarded

HSC-NIJI Blue: Painting the large-scale structure, galaxies, and AGN at cosmic noon

PI: Hironao Miyatake

Intensive program; time shown is the S26A semester allocation.

[Program details ↗](#)

2025

Co-I Cycle 12 2025.1.01077.S

ALMA / Band 3

30.7 h requested

ALMA CO Mapping of the Highest-Density QSO Peak Identified Across a 10,000 deg² Sky

PI: Tomokazu Kiyota · East Asia

Grade C. Conditional queue priority; execution is not guaranteed.

[Program details ↗](#)

PI Cycle 12 2025.1.01243.S

ALMA / Bands 4 & 7

15.8 h requested

Cosmic Himalayas: Mapping Gas and Dust in the Hosts of the Densest QSOs at z~2 with ALMA Bands 4 and 7

PI: Yongming Liang · East Asia

Grade C. Conditional queue priority; execution is not guaranteed.

[Program details ↗](#)

PI Cycle 12 2025.1.01615.S

ALMA / Band 3

14.5 h requested

The Molecular Backbone of the Cosmic Himalayas: Mosaic CO(3-2) Mapping with ALMA Band 3

PI: Yongming Liang · East Asia

Grade C; conditional queue priority, with no guaranteed execution.

PI 2025B 2025B-897187

Blanco 4-m / NEWFIRM

4 nights awarded

How Different Galaxies Interplay with the Densest QSOs at Cosmic Noon?

PI: Yongming Liang · NOIRLab

PI Cycle 27 27700187

Chandra / ACIS-I

340 ks awarded

X-ray Census of Cosmic Himalayas: Super Massive Black Hole Growth at the Cosmic Extremes

PI: Yongming Liang

[Program details ↗](#)

PI 2025B M25BP006

JCMT / SCUBA-2

18 h awarded

Interplay between Galaxies and the Most Extremely Concentrated Quasars

PI: Yongming Liang

Weather band 1.

[Program details ↗](#)

Co-I S25B S25B-066

Keck / KCWI

1 night awarded

Filling the Last Missing Piece in the Giant Ly α Emitter Himiko at $z \sim 7$

PI: Tomokazu Kiyota · Subaru time exchange

Program details ↗

Collaborator 2025B

Keck / KCWI & MOSFIRE

2 nights awarded

Cosmic Himalayas: optical and near-infrared spectroscopy

Program leads: Naveen Reddy / Sebastiano Cantalupo · UC time

Combined collaboration entry: 1 KCWI night and 1 MOSFIRE night for Cosmic Himalayas within the participating programs.

Co-I 2025A

Keck / KCWI

1.5 nights awarded

JWST-ALMA-Keck Synergy Study on the Circumgalactic Gas Accretion in a Giant Ly α Blob

PI: Xiaohui Fan · NASA

Program details ↗

PI 2025B 2025B-643331

Magellan / IMACS

1 night awarded

Unveiling Galaxies and QSOs Within the Densest QSO Region at Cosmic Noon With Magellan/IMACS

PI: Yongming Liang · NOIRLab

Co-I S25B S25B-024

Subaru / PFS-Q

2.45 nights awarded

The diverse cluster evolution probed with HSC2 blue clusters at $0.6 < z < 1.5$

PI: Mariko Kubo

Queue allocation; 3.5 nights including queue overheads.

Program details ↗

2024

Co-I Cycle 11 2024.1.01259.S

ALMA / Bands 1 & 4

Grade C queue

JAKS: JWST-ALMA-Keck Synergy Study on the Circumgalactic Cold Gas Accretion

PI: Haibin Zhang · East Asia

Grade C. Conditional queue priority; execution is not guaranteed.

Program details ↗

PI 2024B 2024B-182778

Blanco 4-m / NEWFIRM

3 nights awarded

How Different Galaxies Interplay with the Densest QSOs at Cosmic Noon?

PI: Yongming Liang · NOIRLab

Co-I Cycle 26 26700289

Chandra / ACIS-I

120 ks awarded

X-ray nuclear activity in an extreme protocluster with the densest population of Lyman alpha nebulae

PI: Yuanyuan Su

[Program details ↗](#)

PI 2024B GN-2024B-FT-204

Gemini North / GNIRS

Approved program

Spectroscopic inspection of luminous SMGs around the densest QSOs at Cosmic Noon

PI: Yongming Liang · Fast Turnaround

[Program details ↗](#)

Co-I Cycle 31 · VLA 2024B joint program HST-17627

Hubble / WFC3

2 orbits awarded

Origin and Feedback of a Giant Radio Galaxy in an Extreme Protocluster at $z=2.3$

PI: Mingyu Li

Joint allocation with VLA/24B-358.

[Program details ↗](#)

PI 2024B M24BP024

JCMT / SCUBA-2

8.7 h awarded

Missing Power of the Largest Group of Enormous Lyman-alpha Nebulae at Cosmic Noon

PI: Yongming Liang

Weather band 1.

[Program details ↗](#)

PI 2024B M24BP052

JCMT / SCUBA-2

13 h awarded

Interplay between Galaxies and the Most Extremely Concentrated Quasars

PI: Yongming Liang

Weather band 2.

[Program details ↗](#)

PI 2024A M24AP004

JCMT / SCUBA-2

8.7 h awarded

Missing Power of the Largest Group of Enormous Lyman-alpha Nebulae at Cosmic Noon

PI: Yongming Liang

Weather band 1.

[Program details ↗](#)

Co-I 2024B MKT-24160

MeerKAT / L band

20 h awarded

MeerKAT follow-up of the large-scale overdense structures containing the most extremely concentrated Quasars at the Cosmic Noon

PI: Fangxia An

Priority B1.

PI Winter 2024 W24DJ

NOEMA / PolyFiX

Approved program

Molecular gas in the luminous SMGs around the densest QSOs at Cosmic Noon

PI: Yongming Liang

[Program details ↗](#)

Co-I S24B S24B-021

Subaru / HSC-Q

1.3 nights awarded

Simultaneous Ly α and H α Observations at $z \sim 6$ with Subaru and JWST

PI: Shunta Shimizu

Queue allocation; 1.9 nights including queue overheads.

[Program details ↗](#)

Co-I S24B S24B-077

Subaru / MOIRCS

2 nights awarded

Mass-dependent Diversity of Protoclusters at $z \sim 4$ detected by Deep-Learning

PI: Yoshihiro Takeda

[Program details ↗](#)

PI S24B S24B-135

Subaru / MOIRCS

2 nights awarded

Interplay between the Most Extremely Concentrated Quasars and Galaxies

PI: Yongming Liang

[Program details ↗](#)

Co-I S24A S24A-057

Subaru / MOIRCS

1 night awarded

Emission-line Mapping in a Protocluster at $z=3$ Fed by Inspiring Cold Streams

PI: Rhythm Shimakawa

[Program details ↗](#)

PI S24A S24A-091

Subaru / HSC-Q & MOIRCS

1.8 nights awarded

Origin and properties of an intergalactic-scale, metal-enriched filament at $z=2.3$

PI: Yongming Liang

MOIRCS: 1.5 nights; HSC queue: 0.3 nights (0.5 nights including queue overheads).

[Program details ↗](#)

Co-I 2024B VLA/24B-358

VLA / L band

9 h awarded

Origin and Feedback of a Giant Radio Galaxy in an Extreme Protocluster at $z=2.3$

PI: Mingyu Li

Maximum approved VLA time; joint HST allocation is listed separately.

[Program details ↗](#)

2023

PI 2023B M23BP028

JCMT / SCUBA-2

12.3 h awarded

Interplay between Galaxies and the Most Extremely Concentrated Quasars

PI: Yongming Liang

Weather band 2.

[Program details ↗](#)

Co-I S23B S23B-054

Keck / MOSFIRE

2 nights awarded

Nature of the quiescent protocluster at $z = 3.0$ revealed by Keck and JWST

PI: Kei Ito · Subaru time exchange

[Program details ↗](#)

Co-I S23B S23B-007

Subaru / FOCAS

3 nights awarded

A Massive Protocluster at $z\sim 3$ Revealed with Light and Shadow

PI: Nobunari Kashikawa

[Program details ↗](#)

Co-I S23B S23B-022

Subaru / HSC-Q

2.4 nights awarded

Subaru meets JWST: Simultaneous measurement of H α and Ly α

PI: Nobunari Kashikawa

[Program details ↗](#)

Co-I S23B S23B-068

Subaru / FOCAS

3 nights awarded

Detecting gas reservoirs around the first rapidly growing SMBHs at $z\sim 6$

PI: Rikako Ishimoto

[Program details ↗](#)

PI

S23B S23B-117

Subaru / MOIRCS

1 night awarded

Interplay between the Most Extremely Concentrated Quasars and Galaxies

PI: Yongming Liang

[Program details ↗](#)**2022****PI**

S22B S22B-059

Subaru / HSC

1 night awarded

Mass assembly revealed in the most massive structures at the Cosmic Noon

PI: Yongming Liang

[Program details ↗](#)**PI**

S22A S22A-093

Subaru / SWIMS

1 night awarded

How different galaxies trace large scale structures with IGM HI at $z \sim 2$?

PI: Yongming Liang

[Program details ↗](#)**2021****Co-I**

Cycle 41 (2021–2022) 41_103

uGMRT / Band 4

39 h observed

Characterising large-scale overdense galaxy structures at $z=2.2$ with deep uGMRT observations

PI: Fangxia An

[Program details ↗](#)**2020****Co-I**

2020B

Magellan / IMACS

2 nights awarded

IGM tomography for a $z \approx 2.2$ large-scale filament associated with a QSO overdensity

PI: Xiaohui Fan

PI

S20A S20A-108

Subaru / HSC

1 night awarded

Galaxy–IGM HI correlation based on $z \sim 2.2$ MAMMOTH overdensities

PI: Yongming Liang

[Program details ↗](#)

2018

Co-I

2018A

Keck / MOSFIRE

1.5 nights awarded

Deep near-IR spectroscopy of the most distant galaxy known at $z=11.1$ II

PI: Nobunari Kashikawa · MWKO

PI

S18B S18B-127

Subaru / HSC(Queue)

1 night awarded

Mapping the most massive overdensity through HI

PI: Yongming Liang

Awarded queue time; observations were not executed.

Program details ↗

Papers & preprints

2026

Jul 2026 Preprint

Subaru meets JWST: A Direct Measurement of Ly α Escape Fraction at $z \approx 6.2$ with Dual Narrow-Band Imaging

Shimizu, Shunta et al.

arXiv:2607.08264

[ADS](#) [DOI](#) [arXiv](#)

Jul 2026

JWST COSMOS-3D: Spectroscopic census and luminosity function of [O III] emitters at $6.75 < z < 9.05$ in COSMOS

Meyer, Romain A. et al.

A&A, 712, A16

[ADS](#) [DOI](#) [arXiv](#)

Jun 2026 Preprint

Prime Focus Spectrograph on the Subaru Telescope: Overview of Science Operations

Tanaka, Masayuki et al.

arXiv:2606.14012

[ADS](#) [DOI](#) [arXiv](#)

Jun 2026

Subaru High- z Exploration of Low-Luminosity Quasars (SHELLQs) — XXV. Large-scale environments of low-luminosity quasars at $z \sim 6$ traced by Ly α emitters

Arita, Junya et al.

MNRAS, 549, stag784

[ADS](#) [DOI](#) [arXiv](#)

Apr 2026

Cosmic Himalayas in CROCODILE: Probing the Extreme Quasar Overdensities by Count-in-cells Analysis and Nearest-neighbor Distribution

Kuwayama, Yuto et al.

ApJ, 1001, 172

[ADS](#) [DOI](#) [arXiv](#)

Mar 2026 [Preprint](#)

A Python-Based Peeling Framework for Radio Interferometry: Application to uGMRT 650MHz Imaging

Peng, Hao et al.

arXiv:2603.10758

[ADS](#) [DOI](#) [arXiv](#)

Feb 2026

Deeper detection limits in astronomical imaging using self-supervised spatiotemporal denoising

Guo, Yuduo et al.

Science, 392, eady9404

[ADS](#) [DOI](#) [arXiv](#)

2025

Dec 2025 [Preprint](#)

Little red dot variability over a century reveals black hole envelope via a giant Einstein cross

Zhang, Zijian et al.

arXiv:2512.05180

[ADS](#) [DOI](#) [arXiv](#)

Jul 2025

RIDEN pilot survey: broad-band selection of candidate quasars with extended Lyman- α nebulae using CLAUDS–HSC–SSP–DUNES² joint data

Shimakawa, Rhythm et al.

MNRAS, 540, 3565–3582

[ADS](#) [DOI](#) [arXiv](#)

Jun 2025

Cosmic Himalayas: The Highest Quasar Density Peak Identified in a 10,000 deg² Sky with Spatial Discrepancies between Galaxies, Quasars, and IGM H I

Liang, Yongming et al.

ApJ, 986, 60

[ADS](#) [DOI](#) [arXiv](#)

Apr 2025

SILVERRUSH. XIV. Ly α Luminosity Functions and Angular Correlation Functions from $\sim 20,000$ Ly α Emitters at $z \sim 2.2\text{--}7.3$ from up to 24 deg^2 HSC-SSP and CHORUS Surveys: Linking the Postreionization Epoch to the Heart of Reionization

Umeda, Hiroya et al.

ApJS, 277, 37

[ADS](#) [DOI](#) [arXiv](#)

Mar 2025

The Physical Origin of Extreme Emission Line Galaxies at High Redshifts: Strong [O III] Emission Lines Produced by Obscured Active Galactic Nuclei

Zhu, Chenghao et al.

ApJ, 982, 27

[ADS](#) [DOI](#) [arXiv](#)

Mar 2025

MAMMOTH-Subaru. IV. Large Scale Structure and Clustering Analysis of Ly α Emitters and Ly α Blobs at $z = 2.2\text{--}2.3$

Zhang, Haibin et al.

ApJ, 981, 70

[ADS](#) [DOI](#) [arXiv](#)

Feb 2025

Predicting Ly α emission from distant galaxies with neural network architecture

Yoshioka, Takehiro et al.

MNRAS, 536, 3386–3400

[ADS](#) [DOI](#) [arXiv](#)

2024

Dec 2024

MAMMOTH-Subaru. II. Diverse Populations of Circumgalactic Ly α Nebulae at Cosmic Noon

Li, Mingyu et al.

ApJS, 275, 27

[ADS](#) [DOI](#) [arXiv](#)

Dec 2024

Mining for Protoclusters at $z \sim 4$ from Photometric Data Sets with Deep Learning

Takeda, Yoshihiro et al.

ApJ, 977, 81

[ADS](#) [DOI](#) [arXiv](#)

Sep 2024

Ly α Imaging around the Hyperluminous Dust-obscured Quasar W2246–0526 at $z = 4.6$

Luo, Yibin et al.

ApJ, 972, 51

[ADS](#) [DOI](#) [arXiv](#)

Jan 2024

MAMMOTH-Subaru. V. Effects of Cosmic Variance on Ly α Luminosity Functions at $z = 2.2\text{--}2.3$

Ma, Ke et al.

ApJ, 961, 102

[ADS](#) [DOI](#) [arXiv](#)

Jan 2024

MAMMOTH-Subaru. III. Ly α Halo Identified by Stacking ~ 3300 Ly α Emitters at $z = 2.2\text{--}2.3$

Zhang, Haibin et al.

ApJ, 961, 63

[ADS](#) [DOI](#) [arXiv](#)

2023

Sep 2023

SILVERRUSH. XIII. A Catalog of 20,567 Ly α Emitters at $z = 2\text{--}7$ Identified in the Full-depth Data of the Subaru/HSC-SSP and CHORUS Surveys

Kikuta, Satoshi et al.

ApJS, 268, 24

[ADS](#) [DOI](#) [arXiv](#)

Sep 2023

Subaru High- z Exploration of Low-luminosity Quasars (SHELLQs). XVIII. The Dark Matter Halo Mass of Quasars at $z \sim 6$

Arita, Junya et al.

ApJ, 954, 210

[ADS](#) [DOI](#) [arXiv](#)

Mar 2023

COSMOS2020: Discovery of a Protocluster of Massive Quiescent Galaxies at $z = 2.77$

Ito, Kei et al.

ApJL, 945, L9

[ADS](#) [DOI](#) [arXiv](#)

2022

Oct 2022

The physical origin for spatially large scatter of IGM opacity at the end of reionization: The IGM Ly α opacity-galaxy density relation

Ishimoto, Rikako et al.

MNRAS, 515, 5914–5926

[ADS](#) [DOI](#) [arXiv](#)

Apr 2022

Third data release of the Hyper Suprime-Cam Subaru Strategic Program

Aihara, Hiroaki et al.

PASJ, 74, 247–272

[ADS](#) [DOI](#) [arXiv](#)

Mar 2022

CHORUS. IV. Mapping the Spatially Inhomogeneous Cosmic Reionization with Subaru HSC

Yoshioka, Takehiro et al.

ApJ, 927, 32

[ADS](#) [DOI](#) [arXiv](#)

Feb 2022

A Wide and Deep Exploration of Radio Galaxies with Subaru HSC (WERGS). VI. Distant Filamentary Structures Pointed Out by High- z Radio Galaxies at $z \sim 4$

Uchiyama, Hisakazu et al.

ApJ, 926, 76

[ADS](#) [DOI](#) [arXiv](#)

2021

Jul 2021

Interrelation of the Environment of Ly α Emitters and Massive Galaxies at $2 < z < 4.5$

Ito, Kei et al.

ApJ, 916, 35

[ADS](#) [DOI](#) [arXiv](#)

Jun 2021

Probing Feedback via IGM tomography and the Ly α Forest with Subaru PFS, TMT/ELT, and JWST

Nagamine, Kentaro et al.

ApJ, 914, 66

[ADS](#) [DOI](#) [arXiv](#)

Apr 2021

SILVERRUSH X: Machine Learning-aided Selection of 9318 LAEs at $z = 2.2, 3.3, 4.9, 5.7, 6.6$, and 7.0 from the HSC SSP and CHORUS Survey Data

Ono, Yoshiaki et al.

ApJ, 911, 78

[ADS](#) [DOI](#) [arXiv](#)

Jan 2021

Evidence for GN-z11 as a luminous galaxy at redshift 10.957

Jiang, Linhua et al.

Nature Astronomy, 5, 256–261

[ADS](#) [DOI](#) [arXiv](#)

Jan 2021

Statistical Correlation between the Distribution of Ly α Emitters and Intergalactic Medium H I at $z \sim 2.2$ Mapped by the Subaru/Hyper Suprime-Cam

Liang, Yongming et al.

ApJ, 907, 3

[ADS](#) [DOI](#) [arXiv](#)

2020

Dec 2020

CHORUS. I. Cosmic HydrOgen Reionization Unveiled with Subaru: Overview

Inoue, Akio K. et al.

PASJ, 72, 101

[ADS](#) [DOI](#) [arXiv](#)

Dec 2020

Faint Quasars Live in the Same Number Density Environments as Lyman Break Galaxies at $z \sim 4$

Uchiyama, Hisakazu et al.

ApJ, 905, 125

[ADS](#) [DOI](#) [arXiv](#)

Nov 2020

Subaru High- z Exploration of Low-luminosity Quasars (SHELLQs). XI. Proximity Zone Analysis for Faint Quasar Spectra at $z \sim 6$

Ishimoto, Rikako et al.

ApJ, 903, 60

[ADS](#) [DOI](#) [arXiv](#)

Aug 2020

The UV Luminosity Function of Protocluster Galaxies at $z \sim 4$: The Bright-end Excess and the Enhanced Star Formation Rate Density

Ito, Kei et al.

ApJ, 899, 5

[ADS](#) [DOI](#) [arXiv](#)

2019

Dec 2019

Second data release of the Hyper Suprime-Cam Subaru Strategic Program

Aihara, Hiroaki et al.

PASJ, 71, 114

[ADS](#) [DOI](#) [arXiv](#)

Jun 2019

The Brightest UV-selected Galaxies in Protoclusters at $z \sim 4$: Ancestors of Brightest Cluster Galaxies?

Ito, Kei et al.

ApJ, 878, 68

[ADS](#) [DOI](#) [arXiv](#)

Jan 2019

Suppression of Low-mass Galaxy Formation around Quasars at $z \sim 2-3$

Uchiyama, Hisakazu et al.

ApJ, 870, 45

[ADS](#) [DOI](#) [arXiv](#)
